# Are They Really Harmless?

## Essential Oil and Aroma Product Exposures Reported to Taiwan Poison Control Center, 2022–2025

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Taipei, Taiwan



**142** Total cases

**73.9%** Age <18 yr

**65.5%** Age 0–5 yr

**88.7%** Oral ingestion

**86.4%** Diffuser → 0–5 yr

## INTRODUCTION

### Background

Essential oils and aroma products are increasingly used in households and are widely perceived as natural and safe. Accidental exposures, particularly among young children, continue to be reported to poison centers. Data describing these exposures in Asian populations remain limited.

### Objective

To characterize essential oil and aroma product exposures reported to PCC-Taiwan, focusing on product categories, age distribution, exposure routes, and clinical presentation.

## METHODS

### Study design

Retrospective review of telephone enquiries on essential oil/aroma exposures reported to PCC-Taiwan during 2022–2025.

### Case identification — keyword search + manual review:

- Search terms: oil, diffuser, aroma, fragrance.

### Product classification

Products were classified according to reported product name, intended use, and available ingredient information.

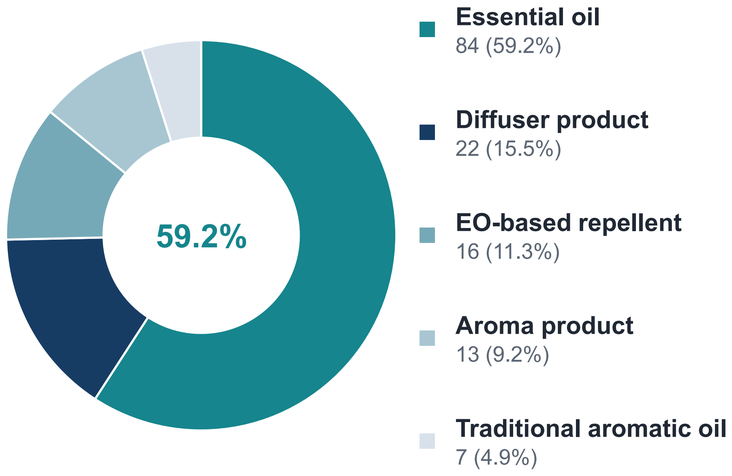
- Essential Oil
- Diffuser Product
- EO-based Repellent
- Aroma Product
- Traditional Aromatic Oil

**ANALYSIS WORKFLOW**

1. **PCC exposure records** — Source database
2. **Keyword screening** — Case identification
3. **Product classification** — 5 product categories
4. **Descriptive analysis** — Age · route · symptoms

**Improved caregiver awareness and safe storage practices may reduce accidental essential oil and aroma product exposures among young children.**

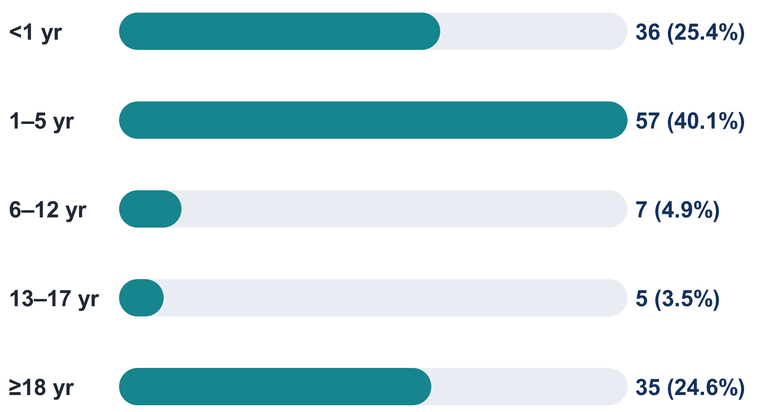
## FIGURE 1. PRODUCT CATEGORIES (N=142)




**Essential oils accounted for 59.2% of reported exposures; diffuser products were the second most common category.**

## FIGURE 2. AGE DISTRIBUTION (N=142)

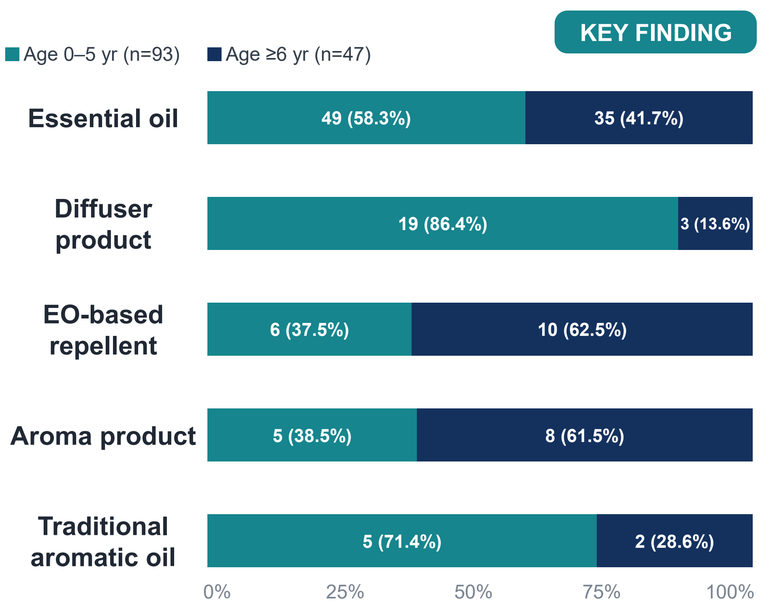
**0–5 yr = 65.5% of all exposures (n=93)**




**Individuals younger than 18 yr represented nearly three-quarters (73.9%) of reported cases. Age was unknown in 2 cases (1.4%).**

## FIGURE 3. PRODUCT CATEGORY × AGE GROUP (N=142)

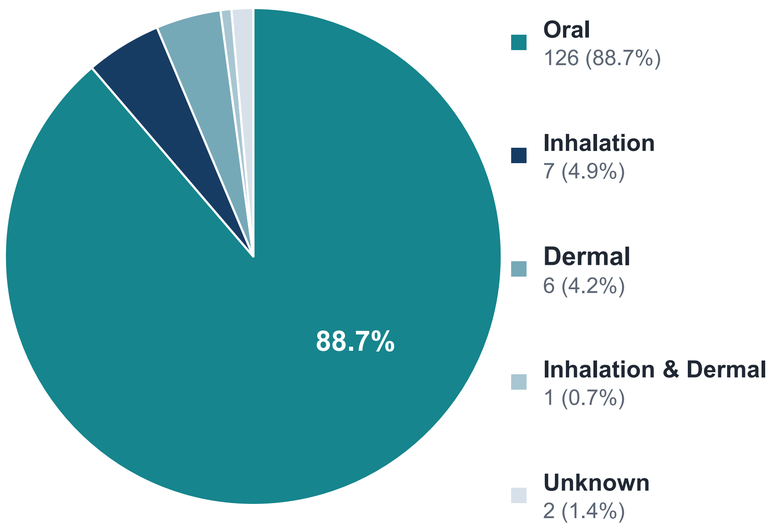
KEY FINDING




**86.4%** 19 of 22 diffuser product exposures occurred among children aged 0–5 years.

**Age patterns varied by product, with diffuser exposures most concentrated in children aged 0–5 years. Two cases with unknown age were excluded from this analysis.**

## FIGURE 4. EXPOSURE ROUTES (N=142)



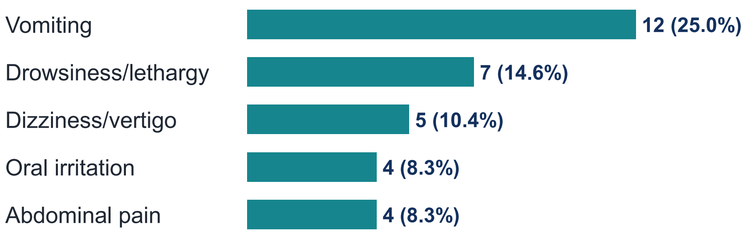

**Oral ingestion was the predominant route, accounting for 88.7% of all cases.**

## CLINICAL MANIFESTATIONS & SEVERITY (N=142)

**63.4%** No symptoms reported (n=90)

**33.8%** Symptoms reported (n=48)

**Most frequently reported symptoms (n=48):**




Symptom status unavailable or indeterminate: 4 (2.8%)

**Clinical outcome severity (N=142):**

| 50.7% | 23.2% | 0.7% | 0.7% |
|---|---|---|---|
| No effect (n=72) | Mild (n=33) | Severe (n=1) | Fatal (n=1) |

Moderate: 0 (0.0%); Not assessed: 29 (20.4%); Unable to determine or unrelated: 6 (4.2%)

## KEY FINDINGS & PUBLIC HEALTH IMPLICATIONS

**Most exposures in young children**
65.5% aged 0–5 yr

**Oral ingestion predominant route**
88.7% of all cases

**No symptoms in most cases**
63.4% of all cases

**Diffuser exposures in young children**
19/22 (86.4%) in children aged 0–5 yr

**Preventable exposures**
Caregiver awareness & safe storage

## STUDY LIMITATIONS

1. Product identification and exposure details relied primarily on caller-reported information.
2. Symptom and outcome assessments were unavailable or incomplete in some cases.
3. Voluntary reporting may underestimate the true number of exposures.

## INTERPRETATION

Exposures occurred predominantly in young children, and most resulted from oral ingestion.
Diffuser products showed a distinct exposure pattern, with 86.4% of cases occurring among children aged 0–5 years.
Most cases had no symptoms reported; nevertheless, these exposures were largely preventable.



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